MILESTONE 4

Data Validation and Error Handling

AI-Based Industrial Maintenance and Process Optimization System

1. Objective

Integrate a common validation layer into single and batch machine-failure prediction so missing, malformed, non-finite, or physically implausible telemetry is rejected before model inference.

2. Delivered implementation

• **src/validation.py:** shared schema, product-type rules, numeric parsing, finite-value checks, configured ranges, and opt-in missing-cell imputation.

• **src/prediction.py:** validation before model loading/inference; clear InvalidInputError for malformed single and batch requests; threshold validation.

• **src/data_validation.py:** compatibility wrapper for CSV validation.

• **src/features.py:** defensive validation before derived feature calculation.

• **Dashboard modules:** single and batch error handling; batch policy rejects invalid values and only imputes missing cells when explicitly enabled.

• **Tests and documentation:** new regression tests and detailed validation scenario report.

3. Required parameters and configured ranges

Parameter	Accepted values
Product Type	L, M, or H
Air temperature [K]	270–330 K
Process temperature [K]	280–340 K
Rotational speed [rpm]	500–4,000 rpm
Torque [Nm]	1–150 Nm
Tool wear [min]	0–400 min

These are broad configured sanity limits for the benchmark telemetry used in this project. They are not universal limits for every industrial machine; update src/validation.py to match actual equipment specifications.

4. Validation policy

• All six feature columns are required; empty input and empty batches are rejected.

• Numeric strings are converted. Invalid text, booleans, NaN, and infinity are rejected.

• Product type is normalized to uppercase and must be L, M, or H.

• Missing cells are rejected by default. Batch missing-cell imputation is opt-in and uses documented nominal defaults.

• Out-of-range values and invalid categories are always rejected; values are never silently clipped.

• If any batch row is invalid, the complete batch is rejected before model loading and prediction.

5. Automated test results

48 tests passed in the focused validation, prediction, integration, and preprocessing suite. The run emitted 7 scikit-learn InconsistentVersionWarning warnings because the model was serialized with version 1.9.0 and the verification environment used version 1.8.0.

Test command

```
python -m pytest -q tests/test_validation.py tests/test_prediction.py tests/test_integration.py
tests/test_preprocessing.py tests/test_milestone4_validation.py
```

Whole-project syntax verification

`python -m py_compile src/*.py dashboard/*.py dashboard/views/*.py tests/test_milestone4_validation.py` — passed.

6. Verification limitation

The full test suite could not be collected in this environment because Streamlit was not installed (ModuleNotFoundError while importing tests/test_deliverables.py). Install all pinned dependencies from requirements.txt and rerun python -m pytest -v, then launch streamlit run dashboard/app.py for final target-environment verification.

7. Completion checklist

■ Missing inputs detected

■ Invalid numerical values handled

■ Incorrect data types handled

■ Numerical ranges validated

■ Understandable errors returned

■ Single prediction validation integrated

■ Batch prediction validation integrated

■ Invalid inputs blocked before model inference

■ Validation scenarios and test results documented

Conclusion

The shared validation layer is integrated with the prediction pipeline. Invalid machine inputs are blocked before inference, batch imputation is explicit, and range violations are rejected instead of silently modified. The focused automated test suite passed 46 tests in the available environment. Final full-suite and live-dashboard verification should be repeated after installing the project dependencies.